FPGA

Synchronous FIFO *Memory Style*

Level-5: FPGA-Oriented Synchronous-Read FIFO

LEVEL-5

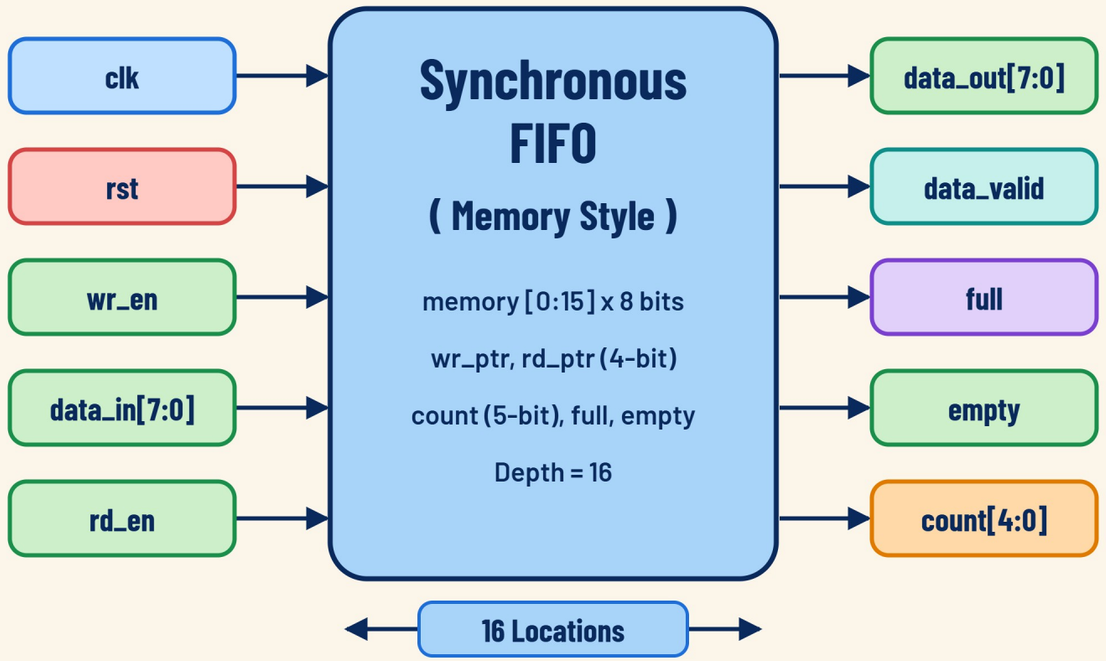
Basic Definition

A **Synchronous FIFO Memory Style** is a parameterized First-In First-Out buffer built on a memory array with a **clocked read**: `data_out` and `data_valid` update on the rising `clk` edge.

Key Features (Level-5)

- ✓ **Parameters:** DATA_WIDTH, DEPTH, ADDR_WIDTH, COUNT_WIDTH
- ✓ **Reset:** synchronous, active-high `rst`
- ✓ **Controls:** independent `wr_en` and `rd_en`
- ✓ **Registered read:** `data_out` + `data_valid`
- ✓ **Pointers:** `wr_ptr`, `rd_ptr` with wraparound
- ✓ **Occupancy:** `count` (0 to 16), `full`, `empty`
- ✓ **Protection:** write rejected if full, read rejected if empty
- ✓ **Simultaneous** read and write supported

Architecture / Block Diagram



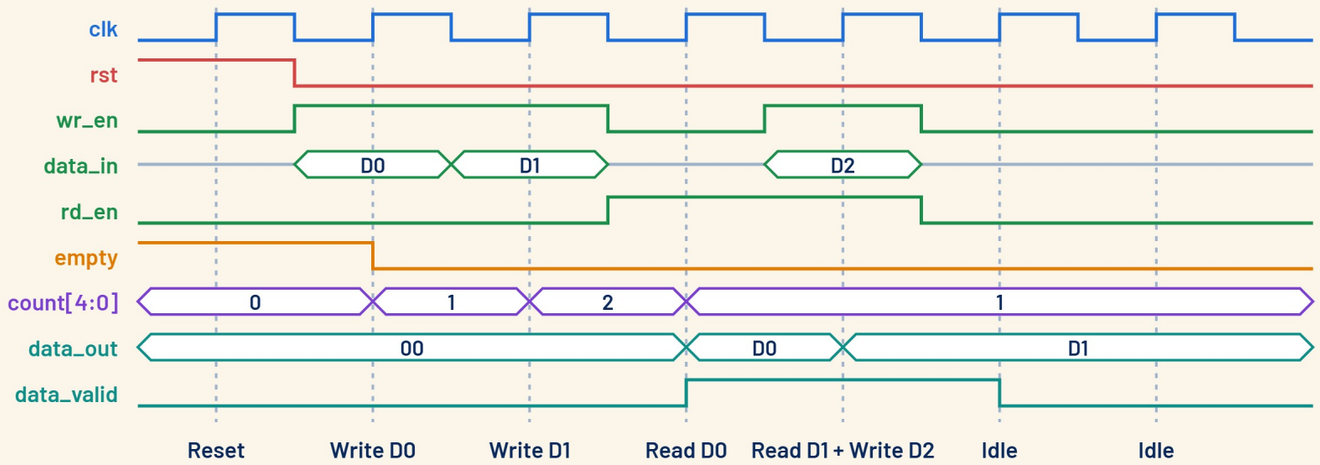
Parameters & Signals

Parameter	Default	Use
`DATA_WIDTH`	8	bits per word
`DEPTH`	16	words stored
`ADDR_WIDTH`	4	pointer bits
`COUNT_WIDTH`	5	counts 0 to 16

How It Works

1. **Write:** if `wr_en` and not full, `data_in` goes to `memory[wr_ptr]`, `wr_ptr` advances, `count` +1.
2. **Read:** if `rd_en` and not empty, `data_out` loads `memory[rd_ptr]`, `data_valid` goes high, `count` -1.
3. **Both:** if a write and a read are accepted on the same clock edge, `count` stays unchanged.

Timing Waveform (Typical Flow)



Important Technical Points

- Read data is registered: updates on the clock edge of an accepted read.
- `data_valid` rises with the new `data_out` word, then drops.
- Rejected reads keep `data_out` unchanged.
- Full when count == 16, empty when count == 0.
- Pointers wrap from DEPTH-1 back to 0.
- Style suits block RAM; confirm in Vivado reports.

Applications

- ✓ Data buffering between modules
- ✓ UART and SPI interfaces
- ✓ Streaming pipelines on FPGA

Key Takeaway

Circular memory, two pointers and a counter, with a registered output built for FPGA block RAM.

Developed by: *Ghulam Yaseen Nangraj*

Verilog-2001 | Vivado 2025.2.1 | Artix-7 Arty A7-100T